

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 9_PAH

Attempt : 2
Total Mark : 30
Marks Obtained : 30

Section 1 : Coding

1. Problem Statement

Arun is building a task manager to keep track of tasks using a LinkedList. The task manager supports the following operations:

"ADD <task>" Adds the given task to the end of the list. "REMOVE" Removes the first task from the list. "SHOW" Displays all tasks in the list in order. If the list is empty, print "EMPTY".

Help Arun implement this functionality using a LinkedList.

Input Format

The first line of the input consists of an integer n, the number of operations.

The next n lines, each containing a command:

- "ADD <task>"
- "REMOVE"
- "SHOW"

Output Format

For each "SHOW" command, the output prints the tasks in order, separated by spaces.

If no tasks exist, print "EMPTY".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

ADD homework

ADD project

SHOW

REMOVE

SHOW

Output: homework project

project

Answer

```
import java.util.*;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int n = sc.nextInt(); // number of operations  
        LinkedList<String> tasks = new LinkedList<>();  
  
        for (int i = 0; i < n; i++) {  
            String command = sc.next(); // read command type  
  
            if (command.equals("ADD")) {  
                String task = sc.next(); // read the task name  
                tasks.add(task);  
            }  
            else if (command.equals("REMOVE")) {  

```


5 10 10 -1 -1 -1

Explanation:

For each element:

4 5 (next greater element)5 102 1010 -1 (No greater element)8 -16 -1

Input Format

The first line contains an integer n, representing the number of elements.

The second line contains n space-separated integers arr[i], where arr[i] is the stock price on the i-th day.

Output Format

The output prints n space-separated integers representing the Next Greater Element for each element in the array.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 6

4 5 2 10 8 6

Output: 5 10 10 -1 -1 -1

Answer

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();           // number of elements
        int[] arr = new int[n];         // input array
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int[] nge = new int[n];         // array to store result
        Stack<Integer> stack = new Stack<>(); // stack to store indexes
```

```

// Traverse the array from right to left
for (int i = n - 1; i >= 0; i--) {
    // Pop all smaller or equal elements
    while (!stack.isEmpty() && stack.peek() <= arr[i]) {
        stack.pop();
    }

    // If stack is empty, no greater element
    if (stack.isEmpty()) {
        nge[i] = -1;
    } else {
        nge[i] = stack.peek(); // top element is next greater
    }

    // Push current element to stack
    stack.push(arr[i]);
}

// Print result
for (int i = 0; i < n; i++) {
    System.out.print(nge[i] + " ");
}
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Rekha is a teacher who wants to calculate the average of marks scored by her students in a test. She needs to store all the marks dynamically because the number of students may vary each time. Using an ArrayList allows her to easily add any number of marks without worrying about the initial size.

Help her implement the task.

Input Format

The first line of input is an integer n , representing the number of students..

The second line of input consists of n double values, representing the marks of each student, separated by a space.

Output Format

The output prints: "Average of the list: " followed by the average value formatted to two decimal places.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

1.0 2.0 3.0 4.0 5.0

Output: Average of the list: 3.00

Answer

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt(); // number of students
        ArrayList<Double> marks = new ArrayList<>(); // dynamic list for marks

        // Read n double values
        for (int i = 0; i < n; i++) {
            marks.add(sc.nextDouble());
        }

        // Calculate sum
        double sum = 0.0;
        for (double m : marks) {
            sum += m;
        }

        // Calculate average
        double avg = sum / n;
```

```
// Print result formatted to 2 decimal places
System.out.printf("Average of the list: %.2f ", avg);

    sc.close();
}
}
```

Status : Correct

Marks : 10/10